

CASE 2 – UI Automation & Dynamic Elements

Objective

The objective of this automation scenario is to validate the secure API key generation behavior on the Developer Portal UI.

Tools Used

- Selenium WebDriver
- Java
- TestNG

Test Scenario

Flow Automated

1. Navigated to the Developer Portal dashboard.
2. Clicked the “Generate New Key” button.
3. Verified that the generated API key is initially displayed in plain text.
4. Validated that after 10 seconds the exact UI element automatically changes into masked format (sk_test_****1234).

Automation Strategy for Stability

UI tests can become unstable due to timing and network fluctuations.

Techniques Used

- Explicit Waits were implemented instead of hardcoded delays.
- Page Object Model (POM) pattern was followed to separate UI locators from test logic.
- Dynamic element synchronization was handled using intelligent wait conditions.

Benefits

- Reduced flaky failures in CI/CD pipelines
- Improved maintainability
- Better readability and reusability of automation scripts